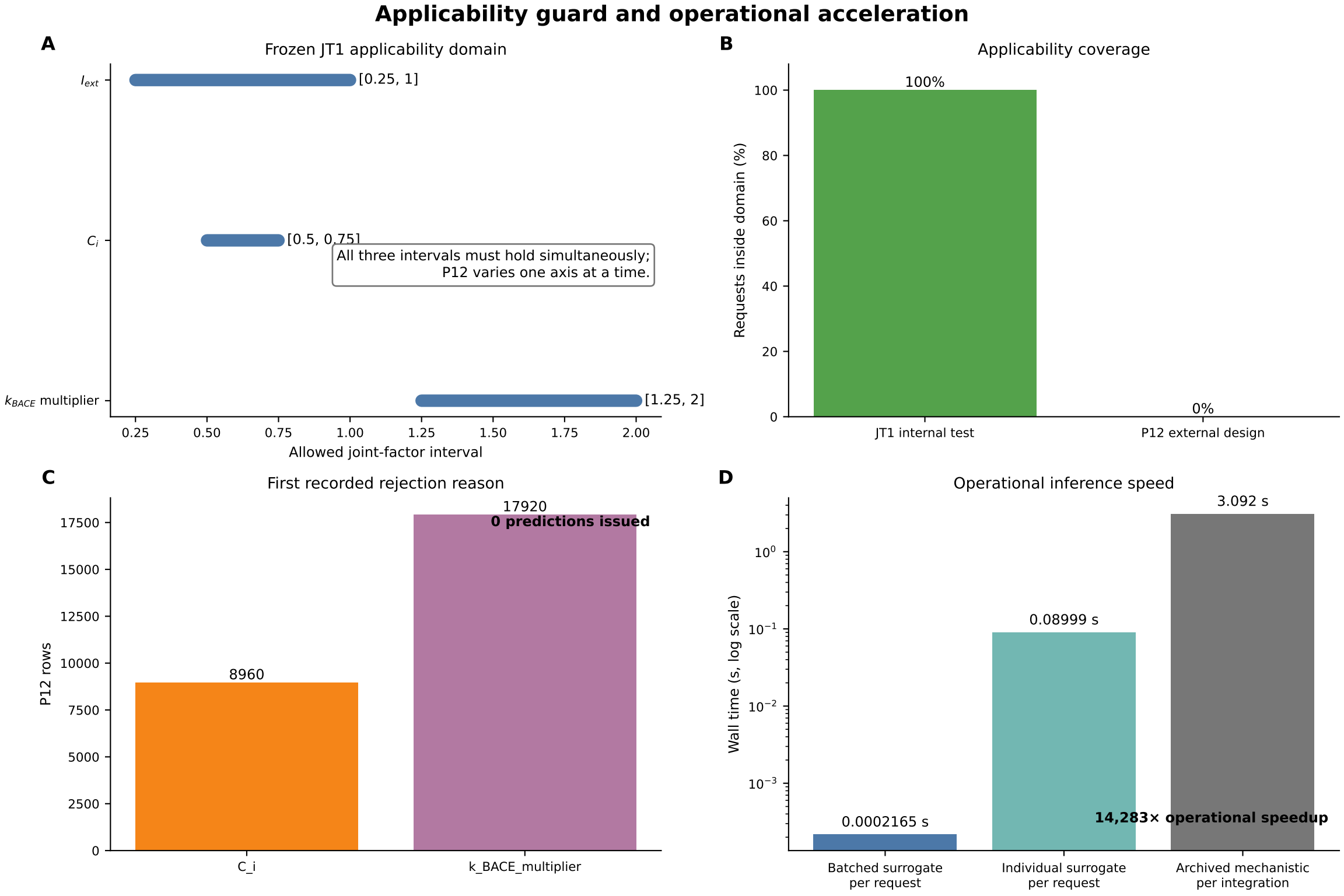
Applicability guard and operational acceleration
A
Frozen JT1 applicability domain
I_{ext}
[0.25, 1]
C_i
[0.5, 0.75]
All three intervals must hold simultaneously;
P12 varies one axis at a time.
k_{BACE} multiplier
[1.25, 2]
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Allowed joint-factor interval
B
Applicability coverage
Requests inside domain (%)
100
80
60
40
20
0
100%
0%
JT1 internal test
P12 external design
C
First recorded rejection reason
P12 rows
17500
15000
12500
10000
7500
5000
2500
0
17920
0 predictions issued
8960
C_i
k_BACE_multiplier
D
Operational inference speed
Wall time (s, log scale)
10^{0}
10^{-1}
10^{-2}
10^{-3}
3.092 s
0.08999 s
0.0002165 s
14,283× operational speedup
Batched surrogate per request
Individual surrogate per request
Archived mechanistic per integration